Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9851747264384045	0.7713404182379157 – 1.0742222222222222

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0052673086097617	0.9046403956437102 – 1.0933024369821536

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9760507399577167	0.8449660678642713 – 1.011104761904762

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9985502367201311	0.9208476303739065 – 1.0949789086950605

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9856326348566568	0.8686482314018108 – 1.0417187028280885

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9230769230769231	0.7752203721841332 – 1.2563076923076921

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9936854234101746	0.9442126930923929 – 1.052763970607238

Independent matrix operands

128 × 128 · 128 × 128 · f32

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9093212885529866	0.8770501260982082 – 0.9883292181069959

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.9583333333333334 – 1.076

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear

CURRENT / BASELINE	1	95% INTERVAL	0.8832 – 1.024

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9997944119201927	0.9500575602961868 – 1.0822704081632653

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0057055889993636	0.9257904545228178 – 1.1766056170546626

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0711008243106563	0.9638131954788399 – 1.282426526539142

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9512175381610398	0.5288740230405842 – 1.398729942304536

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9764277002242503	0.9081814845657666 – 1.4083494301625017

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0443037974683544	0.5116346525718706 – 1.2821479846572525